

Q.19 Classify different types of heat treatment furnace.

Q.20 What is austenite?

Q.21 Define quenching process? Also name few Quenching media?

Q.22 Explain Nitriding process? Also mention few applications?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain Iron Carbon diagram in detail?

Q.24 Explain working principle of heat treatment furnace. What are different selection criteria for heat treatment furnaces?

Q.25 Draw TTT curve? Also explain in brief effect of Alloying addition on TTT diagram?

(40)

(4)

221853B

No. of Printed Pages : 4

Roll No.

221853B

**5th Sem/
Mechanical (Tool & Die Design)
Subject : Heat Treatment**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Lowest critical temperature in iron carbon

- | | |
|----------|-----------|
| a) 727°C | b) 911°C |
| c) 527°C | d) 1150°C |

Q.2 If steel is quenched in oil, the structure obtained will be

- | | |
|--------------|-------------|
| a) Troostite | b) Sorbite |
| c) Bauxite | d) Pearlite |

Q.3 Hyper-eutectoid steel contains carbon

- | | |
|-----------|-----------|
| a) <0.22% | b) <0.77% |
| c) >0.77% | d) None |

(1)

221853B

Q.4 Corrosion resistance of steel is increased by the addition of

- a) Tungsten b) Chromium
- c) Sulphur d) None

Q.5 Which of the following is the hardest?

- a) Pearlite b) Bainite
- c) Martensite d) Austenite

Q.6 Advantages of use of preheated combustion air are

- a) Saving in fuel consumption
- b) Increase in flame temperature
- c) Reduction in scale losses
- d) All of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Process gives maximum hardness to the surface.

Q.8 The instrument used to measure high temperature in the furnace is _____.

Q.9 Ammonia gas is used for the nitriding process (True/False)

Q.10 Pearlite is a combination of _____.

Q.11 Enlist two uses of nitriding.

Q.12 Name any two semi-conductors.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Define creep & fatigue.

Q.14 Write the heat treatment objectives.

Q.15 Differentiate between annealing and hardening.

Q.16 Explain liquid carburizing process?

Q.17 Explain the procedure for flame hardening.

Q.18 Explain various quenching media used in heat treatment.